



Proposal for Capstone Project Design I (EEE 3202)

1.	Project member	:	Labib Marwan Hoque (2101064)
2.	Title of the project	:	Real-Time Edge-Based Facial Recognition System Using Deep Learning on Raspberry Pi 4
3.	Brief description of the project:		
	(a) Project Background:		
	Problem Statement: Biometric security and smart automation are rapidly becoming essential components of modern infrastructure. However, most existing facial recognition systems rely heavily on Cloud Computing, which introduces significant issues regarding data privacy (sending biometric data to external servers), network latency (lag due to internet speed), and high operational costs. Conversely, traditional edge-based solutions (running locally on devices) face a severe hardware bottleneck. Algorithms like Haar Cascades are fast but suffer from poor accuracy in varying lighting or side profiles, while modern Convolutional Neural Networks (CNNs) like Dlib/HOG are accurate but computationally too expensive for cost-effective hardware like the Raspberry Pi, often resulting in unusable frame rates (below 1 FPS). Existing Research & Gap: Current literature explores lightweight models like MobileNet or SqueezeNet, yet implementation on general-purpose single-board computers (SBCs) often requires specialized accelerators (like Google Coral or Nvidia Jetson). There is a distinct lack of optimized software pipelines that can leverage the native CPU of a standard Raspberry Pi 4 to perform Deep Learning inference in real-time without extra hardware. My Approach: This project addresses this gap by implementing a Quantized Deep Learning pipeline using OpenCV's DNN module. We utilize YuNet (a quantized nanometer-scale detector) for face detection and SFace (Sigmoid-Constrained Hypersphere Loss) for recognition. This combination allows us to bypass the computational heaviness of traditional methods, enabling a secure, offline, and real-time biometric system that runs entirely on the edge.		
	(b) Objectives and aims of the project:		
	The primary objective is to develop a standalone, offline facial recognition system. Specific aims include: Objective 1: To implement an optimized Deep Learning pipeline (YuNet + SFace) on the Raspberry Pi 4. Objective 2: To achieve a usable real-time frame rate of 3–5 FPS without external hardware accelerators. Objective 3: To develop a robust "One-Shot" encoding system that allows new users to be registered instantly without re-training the model. Objective 4: To implement dynamic tracking of unregistered individuals ("Unknown" faces) for security monitoring.		
	(c) State how your project address the following program outcomes (PO):		

	POs for EEE 3202	PO Statements
	PO(e) (Modern tool usage)	The project extensively utilizes modern software engineering and computer vision tools. We are applying Python 3.11 for logic, OpenCV 4.6.0 (DNN module) for AI inference, and Picamera2 for low-latency hardware control. Furthermore, we use ONNX (Open Neural Network Exchange) models for platform-agnostic deep learning deployment, demonstrating the application of advanced IT tools to solve complex edge-computing limitations.
	PO(i) (Individual work and teamwork)	As a member/leader of this project, I am responsible for the end-to-end system lifecycle. This involves individual research into model quantization, configuring the Linux environment (Raspberry Pi OS Bookworm), and system integration. If working in a team, roles are divided between hardware setup (cooling/wiring) and software optimization, ensuring effective collaboration in a multidisciplinary setting combining electronics and computer science.
	PO(j) (Communi- cation)	The project involves generating comprehensive design documentation, including block diagrams of the recognition pipeline and performance analysis charts (FPS vs. Accuracy). The system itself communicates visually with the user through a designed UI (Green/Red bounding boxes and confidence scores), and the final outcomes will be presented through a technical poster and report.
	PO(k) (Project manageme- nt and finance)	The project is managed within a strict budget constraint, prioritizing cost-effectiveness by using off-the-shelf components (Pi 4, Standard Camera) rather than expensive industrial sensors. Time management is applied to ensure the software development lifecycle (Design → Train → Test) is completed within the academic timeline.
	(d) Methodology to be adopted in the project:	
	The project follows a linear processing pipeline: Hardware Setup: A Raspberry Pi 4 (4GB) is coupled with a 5MP OV5647 Camera Module via the CSI interface. Active cooling is applied to prevent thermal throttling during AI inference. Image Acquisition: Video is captured at 640×480 resolution using Picamera2 in RGB888 format to ensure color accuracy for the AI models. Pre-processing: Frames are mirrored (flipped) for user intuition and resized to 320×320 to match the input tensor requirements of the neural network. Inference Engine: Detection: The frame is passed to YuNet, which outputs bounding boxes and 5 facial landmarks. Recognition: Detected faces are aligned and passed to SFace, which generates a 128-dimensional embedding vector. Matching Logic: The system calculates Cosine Similarity between live embeddings and a stored .pickle database. A threshold of 0.363 determines if a face is "Known" or "Unknown".	
	(e) Expected results:	
	Performance: A stable frame rate of 3–5 FPS on the CPU, which is sufficient for access control scenarios (e.g., walking up to a door). Accuracy: Greater than 98% accuracy in identifying registered users, with robust performance in low-light and side-profile views (up to ±30°±30°).	

	• Functionality: Successful distinction between multiple registered users and tracking of unregistered intruders simultaneously.			
	(f) Relevance of the project to national development:			
	This project supports the " Digital Bangladesh " and " Smart Bangladesh 2041 " initiatives by providing a low-cost, domestically producible solution for security and digitization. It can be deployed for automated attendance in schools, secure entry in government offices, and low-cost surveillance for small businesses, reducing reliance on expensive imported security solutions.			
4.	Basic facilities available in the department for the proposed investigation:			
	• Computer Laboratories for coding and simulation. • Basic electronics equipment (Power supplies, Monitors, Peripherals) required to interface with the Raspberry Pi. • Library access for research papers on Deep Learning quantization.			
5.	Key performance indicators of the project:			
	1. Frames Per Second (FPS): Target ≥ 3 FPS 2. Inference Latency: Target < 300 ms per frame. 3. True Positive Rate (Accuracy): Target $> 95\%$. 4. False Acceptance Rate (FAR): Target $< 1\%$ (Critical for security).			
6.	Cost Estimation:			
	Sl. No.	Item Description		Cost
		Name of the Item	Quantity	Unit Price (BDT) Total Cost (BDT)
	1.	Raspberry Pi 4 Model B (4GB RAM)	1	12,850 12,850
	2.	Raspberry Pi Camera Module (5MP)	1	880 880
	3.	Micro SD Card (32GB)	1	600 600
	4.	Power Bank	1	2000 2000
	5.	Cable (USB to USB-C)	1	150 150
	6.	Aluminum Heat Sink Set with Adhesive	1	120 120
				Total Expense (Taka) 16,600

Project Member:	Supervisor:
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